03-01-2026 Weekly Assignment 1

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SQL and PL/SQL

Q1. Write CREATE TABLE commands for the following table descriptions: **[15]**

CLIENT_MASTER

CLIENTNO	vc2(6)	PK & Must start with 'C'
NAME	vc2(20)	NOT NULL
ADDRESS1	vc2(30)	
ADDRESS1	vc2(30)	
CITY	vc2(15)	
PINCODE	n(8)	
STATE	vc2(15)	
BALDUE	n (10,2)	

PRODUCT_MASTER

PRODUCTNO	vc2(6)	PK & Must start with 'P'
DESCRIPTION	vc2(15)	NOT NULL
PROFITPERC	n(4,2)	NOT NULL
UNITMEASURE	vc2(10)	NOT NULL
QTYONHAND	n (8)	NOT NULL
REORDERLVL	n(8)	NOT NULL
SELLPRICE	n (8,2)	NOT NULL & can't be 0
COSTPRICE	n (8,2)	NOT NULL & can't be 0

SALESMAN_MASTER

SALESMANNO	vc2(6)	PK & Must start with 'S'
SALESMANNAME	vc2(20)	NOT NULL
ADDRESS1	vc2(30)	NOT NULL
ADDRESS2	vc2(30)	
CITY	vc2(20)	
PINCODE	n(8)	
State	vc2(20)	
SALAMT	n(8,2)	NOT NULL & Can't be 0
TGTTOGET	n(6,2)	NOT NUL
YTDSALES	n(6,2)	NOT NULL
REMARKS	vc2(60)	

SALES_ORDER

ORDERNO	vc2(6)	PK & Must start with 'O'
CLIENTNO	vc2(6)	FK to CLIENT_MASTER
ORDERDATE	date	
DELYADDR	vc2(25)	
SALESMANNO	vc2(6)	FK to SALESMAN_MASTER
DELYTYPE	char(1)	Check 'P' or 'F'
BILLEDYN	char(1)	Check 'Y' or 'N'
DELYDATE	date	
ORDERSTATUS	vc2(10)	Check 'In Process' or 'Fulfilled' or 'Backorder' or 'Cancelled'

SALES_ORDER_DETAILS

ORDERNO	vc2(6)	FK to SALES_ORDER
PRODUCTNO	vc2(6)	FK to PRODUCT_MASTER
QTYORDERED	n(8)	
QTYDISP	n(8)	
PRODUCTRATE	n(10,2)	

Create Composite PRIMARY KEY of ORDERNO and PRODUCTNO

1. DATABASE CREATION

```
CREATE DATABASE sales_db;
USE sales_db;
```

2. TABLE CREATION

CLIENT_MASTER

```
CREATE TABLE Client_Master (
   ClientNo   VARCHAR(6) PRIMARY KEY,
   Name       VARCHAR(20) NOT NULL,
   Address1   VARCHAR(30),
   Address2   VARCHAR(30),
   City       VARCHAR(15),
   Pincode    INT,
   State      VARCHAR(15),
   BalDue     DECIMAL(10,2)
);
```

Output:

partner.sale...Client_Master | SQLQuery5.s... KUMAR (69))*

Column Name	Data Type	Allow Nulls
ClientNo	varchar(6)	☐
Name	varchar(20)	☐
Address1	varchar(30)	☑
Address2	varchar(30)	☑
City	varchar(15)	☑
Pincode	int	☑
State	varchar(15)	☑
BalDue	decimal(10, 2)	☑
		☐

PRODUCT_MASTER

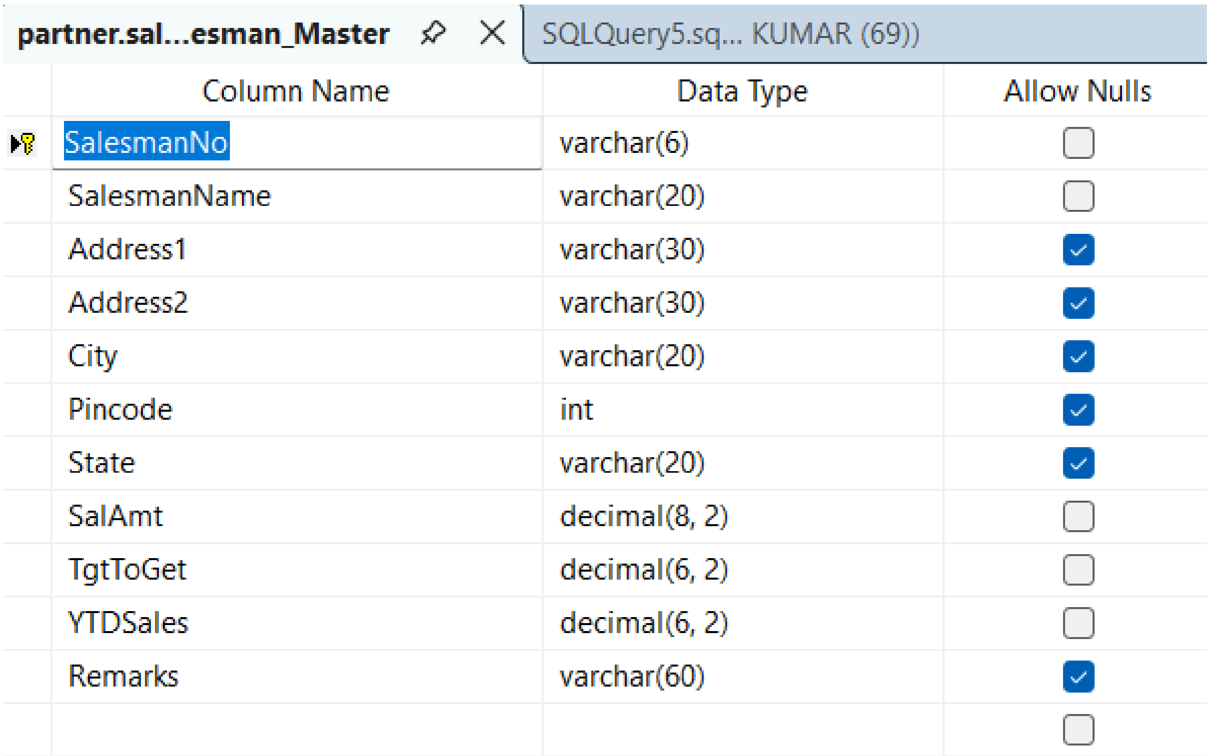
```
CREATE TABLE Product_Master (
    ProductNo      VARCHAR(6) PRIMARY KEY,
    Description    VARCHAR(20) NOT NULL,
    ProfitPercent  DECIMAL(6,2) NOT NULL,
    UnitMeasure    VARCHAR(10) NOT NULL,
    QtyOnHand      INT NOT NULL,
    ReorderLevel   INT NOT NULL,
    SellPrice      DECIMAL(8,2) NOT NULL,
    CostPrice      DECIMAL(8,2) NOT NULL
);
```

partner.sale...oduct_Master | partner.sales_....Client_Master | SQLQuery5.s

Column Name	Data Type	Allow Nulls
ProductNo	varchar(6)	☐
Description	varchar(20)	☐
ProfitPercent	decimal(6, 2)	☐
UnitMeasure	varchar(10)	☐
QtyOnHand	int	☐
ReorderLevel	int	☐
SellPrice	decimal(8, 2)	☐
CostPrice	decimal(8, 2)	☐
		☐

SALESMAN_MASTER

```
CREATE TABLE Salesman_Master (
    SalesmanNo   VARCHAR(6) PRIMARY KEY,
    SalesmanName VARCHAR(20) NOT NULL,
    Address1     VARCHAR(30),
    Address2     VARCHAR(30),
    City         VARCHAR(20),
    Pincode      INT,
    State        VARCHAR(20),
    SalAmt       DECIMAL(8,2) NOT NULL,
    TgtToGet     DECIMAL(6,2) NOT NULL,
    YTDSales     DECIMAL(6,2) NOT NULL,
    Remarks      VARCHAR(60)
);
```



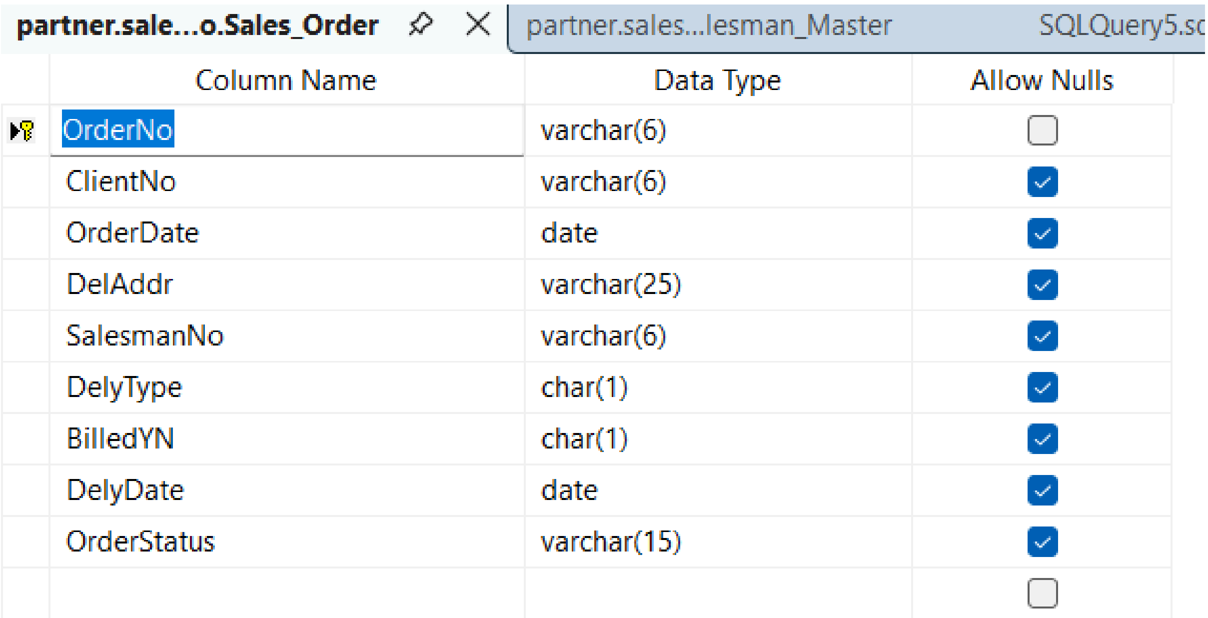
partner.sal...esman_Master | SQLQuery5.sq... KUMAR (69))

Column Name	Data Type	Allow Nulls
SalesmanNo	varchar(6)	☐
SalesmanName	varchar(20)	☐
Address1	varchar(30)	☑
Address2	varchar(30)	☑
City	varchar(20)	☑
Pincode	int	☑
State	varchar(20)	☑
SalAmt	decimal(8, 2)	☐
TgtToGet	decimal(6, 2)	☐
YTDSales	decimal(6, 2)	☐
Remarks	varchar(60)	☑
		☐

SALES_ORDER

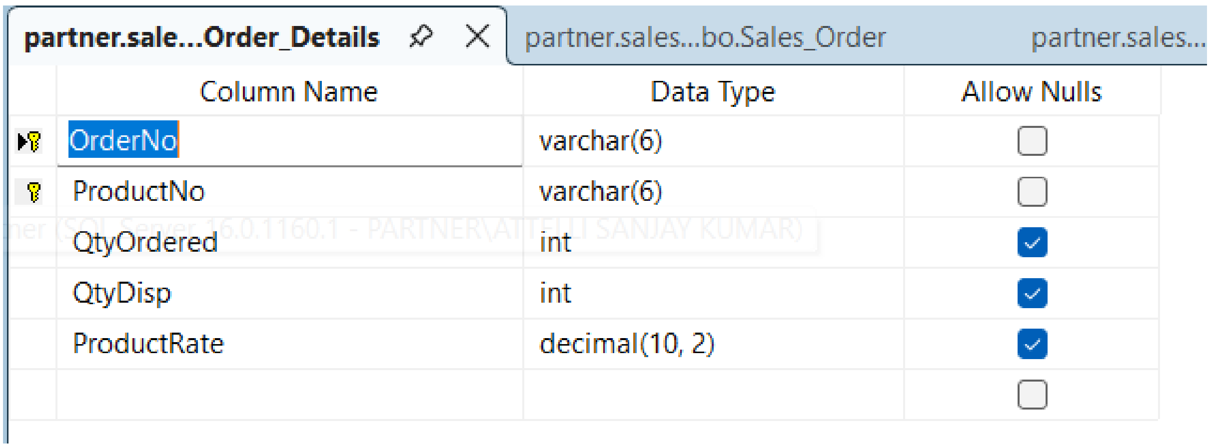
```
CREATE TABLE Sales_Order (
    OrderNo     VARCHAR(6) PRIMARY KEY,
    ClientNo    VARCHAR(6),
    OrderDate   DATE,
    DelAddr     VARCHAR(25),
    SalesmanNo  VARCHAR(6),
    DelyType    CHAR(1) CHECK (DelyType IN ('P','F')),
    BilledYN    CHAR(1) CHECK (BilledYN IN ('Y','N')),
    DelyDate    DATE,
```

```
    OrderStatus VARCHAR(15),

    FOREIGN KEY (ClientNo) REFERENCES Client_Master(ClientNo),
    FOREIGN KEY (SalesmanNo) REFERENCES Salesman_Master(SalesmanNo)
);
```



SALES_ORDER_DETAILS

```
CREATE TABLE Sales_Order_Details (
    OrderNo     VARCHAR(6),
    ProductNo   VARCHAR(6),
    QtyOrdered  INT,
    QtyDisp     INT,
    ProductRate DECIMAL(10,2),

    PRIMARY KEY (OrderNo, ProductNo),
    FOREIGN KEY (OrderNo) REFERENCES Sales_Order(OrderNo),
    FOREIGN KEY (ProductNo) REFERENCES Product_Master(ProductNo)
);
```



Sample Insert Commands

INSERT INTO Client_Master (ClientNo, Name, City, PinCode, State, BalDue) VALUES ('C00001', 'Ivan Bayross', 'Mumbai', 400054, 'Maharashtra', 15000);
INSERT INTO Product_Master VALUES ('P00001', 'T-Shirts', 5, 'Piece', 200, 50, 350, 250);
INSERT INTO Salesman_Master VALUES ('S00001', 'Aman', 'A/14', 'Worli', 'Mumbai', 400002, 'Maharashtra', 3000, 100, 50, 'Good');
INSERT INTO Sales_Order (OrderNo, OrderDate, ClientNo, DelyType, BilledYn, SalesmanNo, DelyDate, OrderStatus)
VALUES('O19001', '12-june-02', 'C00001', 'F', ' N', 'S00001', '20-july-02', 'In Process');
INSERT INTO Sales_Order_Details (OrderNo, ProductNo, QtyOrdered, QtyDisp, ProductRate) VALUES('O19001', 'P00001', 4, 4, 525);

3. INSERT DATA (INDIAN NAMES)

CLIENT_MASTER DATA

INSERT INTO Client_Master VALUES
('C00001','Sanjay','Ameerpet',NULL,'Hyderabad',500016,'Telangana',12000),
('C00002','Manoj','BTM',NULL,'Bangalore',560029,'Karnataka',8000),
('C00003','Varshith','Gachibowli',NULL,'Hyderabad',500032,'Telangana',15000),
('C00004','Shaaz','Andheri',NULL,'Mumbai',400053,'Maharashtra',6000),
('C00005','Nanda','Velachery',NULL,'Chennai',600042,'Tamil Nadu',10000);

Results | Messages

	ClientNo	Name	Address1	Address2	City	Pincode	State	BalDue
1	C00001	Sanjay	Ameerpet	NULL	Hyderabad	500016	Telangana	12000.00
2	C00002	Manoj	BTM	NULL	Bangalore	560029	Karnataka	8000.00
3	C00003	Varshith	Gachibowli	NULL	Hyderabad	500032	Telangana	15000.00
4	C00004	Shaaz	Andheri	NULL	Mumbai	400053	Maharashtra	6000.00
5	C00005	Nanda	Velachery	NULL	Chennai	600042	Tamil Nadu	10000.00

PRODUCT_MASTER DATA

INSERT INTO Product_Master VALUES
('P00001','Laptop',10,'Piece',50,10,60000,54000),
('P00002','Mobile',8,'Piece',80,20,30000,27600),
('P00003','Headphones',12,'Piece',150,30,3000,2640),
('P00004','Keyboard',15,'Piece',100,25,2000,1700);

Results | Messages

	ProductNo	Description	ProfitPercent	UnitMeasure	QtyOnHand	ReorderLevel	SellPrice	CostPrice
1	P00001	Laptop	10.00	Piece	50	10	60000.00	54000.00
2	P00002	Mobile	8.00	Piece	80	20	30000.00	27600.00
3	P00003	Headphones	12.00	Piece	150	30	3000.00	2640.00
4	P00004	Keyboard	15.00	Piece	100	25	2000.00	1700.00

SALESMAN_MASTER DATA

INSERT INTO Salesman_Master VALUES
('S00001','Ritwhik','Madhapur',NULL,'Hyderabad',500081,'Telangana',25000,50000,30000,'Good'),
('S00002','Nanda','Whitefield',NULL,'Bangalore',560066,'Karnataka',22000,45000,28000,'Average');

	SalesmanNo	SalesmanName	Address1	Address2	City	Pincode	State	SalAmt	TgtToGet	YTDSales	Remarks
1	S00001	Ritwhik	Madhapur	NULL	Hyderabad	500081	Telangana	2500.00	5000.00	3000.00	Good
2	S00002	Nanda	Whitefield	NULL	Bangalore	560066	Karnataka	2200.00	4500.00	2800.00	Average

SALES_ORDER DATA

INSERT INTO Sales_Order VALUES
('O19001','C00001','2024-06-10','Hyderabad','S00001','P','N','2024-06-20','In Process'),
('O19002','C00002','2024-06-12','Bangalore','S00002','F','Y','2024-06-18','Fulfilled');

Results | Messages

	OrderNo	ClientNo	OrderDate	DelAddr	SalesmanNo	DelyType	BilledYN	DelyDate	OrderStatus
1	O19001	C00001	2024-06-10	Hyderabad	S00001	P	N	2024-06-20	In Process
2	O19002	C00002	2024-06-12	Bangalore	S00002	F	Y	2024-06-18	Fulfilled

SALES_ORDER_DETAILS DATA

INSERT INTO Sales_Order_Details VALUES
('O19001','P00001',1,1,60000),
('O19001','P00003',2,2,3000),
('O19002','P00002',1,1,30000);

Results | Messages

	OrderNo	ClientNo	OrderDate	DelAddr	SalesmanNo	DelyType	BilledYN	DelyDate	OrderStatus
1	O19001	C00001	2024-06-10	Hyderabad	S00001	P	N	2024-06-20	In Process
2	O19002	C00002	2024-06-12	Bangalore	S00002	F	Y	2024-06-18	Fulfilled

Answer following queries with the help of above schema : [10]

1. Display the names of all the clients.
2. Display all the clients who are located in Mumbai.
3. Display all the products whose selling price is > 2000 and < 5000.
4. Display Name, City and State of Clients not in the state of Maharashtra.
5. Display all the information of client_no C0001 and C0002.
6. Change the selling price of '1.44 drive' to Rs. 1150.50.
7. Delete the record of client_no C0005.
8. Display the clients who stay in a city whose second letter is 'a'.
9. Count the number of products having price greater than or equal to 1500.
10. Display qtyordered, qtydisp and balancedqty (not in table).

Write Commands to do following [10]

1. Make Client_no as primary key in client_master.
2. Add a new column phone_no in the client_master table.
3. Add the not null constraint in the product_master table with the column description, profit percent, sell price and cost price.
4. Change size of name column to 60 in client_master table.
5. Remove pincode column from table.

Define in 1 or 2 lines and give one example also [10]

1. Recursive Relationship.
2. Composite key.
3. The 'like' operator with pattern matching.
4. Drop Table command.
5. Full Outer Join.

Write queries for following descriptions: (Joins) [10]

1. Find out the products, which have been sold to 'Ivan Bayross'.
2. Finding out the products and their quantities that will have to be delivered in the current month.
3. Listing the ProductNo and description of constantly sold (i.e. rapidly moving) products.
4. Finding the names of clients who have purchased 'Trousers'.
5. Listing the products and orders from customers who have ordered less than 5 units of 'Pull Overs'.

Write queries for following descriptions: (Subqueries) [12]

1. Finding the non-moving products i.e. products not being sold.
2. Finding the name and complete address for the customer who has placed Order number 'O19001'.
3. Finding the clients who have placed orders before the month of May'02.

Write Commands to do following [12]

1. Display system date as Saturday, February 11, 2012
2. Display Balance Due from Client master as $99,999.99
3. Display message as 'Salesman Aman sold goods of 50 while given target was 100.
4. Display your Age in Years

4. ANSWER FOLLOWING QUERIES

1. Display the names of all the clients

SELECT Name
FROM Client_Master;

Results Messages

	Name
1	Sanjay
2	Manoj
3	Varshith
4	Shaaz
5	Nanda

2. Display all the clients who are located in Mumbai

SELECT *
FROM Client_Master
WHERE City = 'Mumbai';

Results Messages

	ClientNo	Name	Address1	Address2	City	Pincode	State	BalDue
1	C00004	Shaaz	Andheri	NULL	Mumbai	400053	Maharashtra	6000.00

3. Display all the products whose selling price is > 2000 and < 5000

SELECT *
FROM Product_Master
WHERE SellPrice > 2000
AND SellPrice < 5000;

Results Messages

	ProductNo	Description	ProfitPercent	UnitMeasure	QtyOnHand	ReorderLevel	SellPrice	CostPrice
1	P00003	Headphones	12.00	Piece	150	30	3000.00	2640.00

4. Display Name, City and State of Clients not in the state of Maharashtra

SELECT Name, City, State
FROM Client_Master
WHERE State <> 'Maharashtra';

Results | Messages

	Name	City	State
1	Sanjay	Hyderabad	Telangana
2	Manoj	Bangalore	Karnataka
3	Varshith	Hyderabad	Telangana
4	Nanda	Chennai	Tamil Nadu

5. Display all the information of client no C00001 and C00002

SELECT *
FROM Client_Master
WHERE ClientNo IN ('C00001','C00002');

	ClientNo	Name	Address1	Address2	City	Pincode	State	BalDue
1	C00001	Sanjay	Ameerpet	NULL	Hyderabad	500016	Telangana	12000.00
2	C00002	Manoj	BTM	NULL	Bangalore	560029	Karnataka	8000.00

6. Change the selling price of '1.44 drive' to Rs. 1150.50

UPDATE Product_Master
SET SellPrice = 1150.50
WHERE Description = '1.44 drive';

7. Delete the record of client_no C00005

DELETE FROM Client_Master
WHERE ClientNo = 'C00005';

Messages

```
(1 row affected)

Completion time: 2026-01-04T09:09:22.0548866+05:30
```

Results | Messages

	ClientNo	Name	Address1	Address2	City	Pincode	State	BalDue
1	C00001	Sanjay	Ameerpet	NULL	Hyderabad	500016	Telangana	12000.00
2	C00002	Manoj	BTM	NULL	Bangalore	560029	Karnataka	8000.00
3	C00003	Varshith	Gachibowli	NULL	Hyderabad	500032	Telangana	15000.00
4	C00004	Shaaz	Andheri	NULL	Mumbai	400053	Maharashtra	6000.00

8. Display the clients who stay in a city whose second letter is 'a'

SELECT *
FROM Client_Master
WHERE City LIKE '_a%';

	ClientNo	Name	Address1	Address2	City	Pincode	State	BalDue
1	C00002	Manoj	BTM	NULL	Bangalore	560029	Karnataka	8000.00

9. Count the number of products having price greater than or equal to 1500

```
SELECT COUNT(*) AS Product_Count
FROM Product_Master
WHERE SellPrice >= 1500;
```

	Product_Count
1	4

10. Display Qtyordered, Qtydisp and Balance Qty (not in table)

```
SELECT QtyOrdered,
    QtyDisp,
    (QtyOrdered - QtyDisp) AS BalanceQty
FROM Sales_Order_Details;
```

	QtyOrdered	QtyDisp	BalanceQty
1	1	1	0
2	2	2	0
3	1	1	0

Write Commands to do following [10]

1. Make Client_no as primary key in client_master

ALTER TABLE Client_Master
ADD PRIMARY KEY (ClientNo);

partner.sales_....Client_Master SQLQuery5.s... KUMAR (69))*

Column Name	Data Type	Allow Nulls
ClientNo	varchar(6)	[ ]
Name	varchar(20)	[ ]
Address1	varchar(30)	[x]
Address2	varchar(30)	[x]
City	varchar(15)	[x]
Pincode	int	[x]
State	varchar(15)	[x]
BalDue	decimal(10, 2)	[x]
		[ ]

2. Add a new column phone_no in the client_master table

ALTER TABLE Client_Master
ADD Phone_No VARCHAR(15);

Results Messages

	ClientNo	Name	Address1	Address2	City	Pincode	State	BalDue	Phone_No
1	C00001	Ivan Bayross	Andheri	NULL	Mumbai	400053	Maharashtra	5000.00	NULL
2	C00002	Sanjay	Ameerpet	NULL	Hyderabad	500016	Telangana	8000.00	NULL
3	C00003	Manoj	BTM	NULL	Bangalore	560029	Karnataka	3000.00	NULL
4	C00004	Varshith	Gachibowli	NULL	Hyderabad	500032	Telangana	12000.00	NULL
5	C00005	Shaaz	Velachery	NULL	Chennai	600042	Tamil Nadu	2000.00	NULL
6	C00006	Nanda	Kothrud	NULL	Pune	411038	Maharashtra	9000.00	NULL
7	C00007	Ramesh	Ashok Nagar	NULL	Delhi	110001	Delhi	4000.00	NULL
8	C00008	Suresh	Alkapuri	NULL	Vadodara	390007	Gujarat	6000.00	NULL
9	C00009	Mahesh	Navrangpura	NULL	Ahmedabad	380009	Gujarat	3500.00	NULL
10	C00010	Kiran	Banashankari	NULL	Bangalore	560070	Karnataka	7200.00	NULL
11	C00011	Prakash	Karve Nagar	NULL	Pune	411052	Maharashtra	9100.00	NULL

3. Add the not null constraint in the product_master table

ALTER TABLE Product_Master
ALTER COLUMN Description VARCHAR(20) NOT NULL;
select * from Product_Master

Results Messages

	ProductNo	Description	ProfitPercent	UnitMeasure	QtyOnHand	ReorderLevel	SellPrice	CostPrice
1	P00001	Laptop	10.00	Piece	50	10	60000.00	54000.00
2	P00002	Mobile	8.00	Piece	80	20	30000.00	27600.00
3	P00003	Headphones	12.00	Piece	150	30	3000.00	2640.00
4	P00004	Keyboard	15.00	Piece	100	25	2000.00	1700.00

ALTER TABLE Product_Master
ALTER COLUMN ProfitPercent DECIMAL(6,2) NOT NULL;
select * from Product_Master

Results | Messages

	ProductNo	Description	ProfitPercent	UnitMeasure	QtyOnHand	ReorderLevel	SellPrice	CostPrice
1	P00001	Laptop	10.00	Piece	50	10	60000.00	54000.00
2	P00002	Mobile	8.00	Piece	80	20	30000.00	27600.00
3	P00003	Headphones	12.00	Piece	150	30	3000.00	2640.00
4	P00004	Keyboard	15.00	Piece	100	25	2000.00	1700.00

ALTER TABLE Product_Master
ALTER COLUMN SellPrice DECIMAL(8,2) NOT NULL;
select * from Product_Master

Results | Messages

	ProductNo	Description	ProfitPercent	UnitMeasure	QtyOnHand	ReorderLevel	SellPrice	CostPrice
1	P00001	Laptop	10.00	Piece	50	10	60000.00	54000.00
2	P00002	Mobile	8.00	Piece	80	20	30000.00	27600.00
3	P00003	Headphones	12.00	Piece	150	30	3000.00	2640.00
4	P00004	Keyboard	15.00	Piece	100	25	2000.00	1700.00

ALTER TABLE Product_Master
ALTER COLUMN CostPrice DECIMAL(8,2) NOT NULL;
select * from Product_Master

Results | Messages

	ProductNo	Description	ProfitPercent	UnitMeasure	QtyOnHand	ReorderLevel	SellPrice	CostPrice
1	P00001	Laptop	10.00	Piece	50	10	60000.00	54000.00
2	P00002	Mobile	8.00	Piece	80	20	30000.00	27600.00
3	P00003	Headphones	12.00	Piece	150	30	3000.00	2640.00
4	P00004	Keyboard	15.00	Piece	100	25	2000.00	1700.00

4. Change size of name column to 60 in client_master table

ALTER TABLE Client_Master
ALTER COLUMN Name VARCHAR(60);
select * from Client_Master

Results | Messages

	ClientNo	Name	Address1	Address2	City	Pincode	State	BalDue	Phone_No
1	C00001	Sanjay	Ameerpet	NULL	Hyderabad	500016	Telangana	12000.00	NULL
2	C00002	Manoj	BTM	NULL	Bangalore	560029	Karnataka	8000.00	NULL
3	C00003	Varshith	Gachibowli	NULL	Hyderabad	500032	Telangana	15000.00	NULL
4	C00004	Shaaz	Andheri	NULL	Mumbai	400053	Maharashtra	6000.00	NULL

5. Remove pincode column from table

ALTER TABLE Client_Master
DROP COLUMN Pincode;

	ClientNo	Name	Address1	Address2	City	State	BalDue	Phone_No
1	C00001	Sanjay	Ameerpet	NULL	Hyderabad	Telangana	12000.00	NULL
2	C00002	Manoj	BTM	NULL	Bangalore	Karnataka	8000.00	NULL
3	C00003	Varshith	Gachibowli	NULL	Hyderabad	Telangana	15000.00	NULL
4	C00004	Shaaz	Andheri	NULL	Mumbai	Maharashtra	6000.00	NULL

Define in 1 or 2 lines and Give one example also

1. Recursive Relationship

Definition:
A recursive relationship is when a table is connected to itself, usually to show a hierarchy like employee manager.

Real-time example:
In a company, every employee has a manager, and the manager is also an employee in the same table.

Example Query:

```
SELECT e1.emp_name AS Employee, e2.emp_name AS Manager
FROM employee e1
JOIN employee e2
ON e1.manager_id = e2.emp_id;
```

2. Composite Key

Definition:
A composite key is a primary key made by combining two or more columns to uniquely identify a record.

Real-time example:
In an order system, one order can have many products. So OrderNo alone is not enough OrderNo and ProductNo together identify each record.

Example Query:

```
CREATE TABLE Order_Details (
   OrderNo INT,
   ProductNo INT,
   Quantity INT,
   PRIMARY KEY (OrderNo, ProductNo)
);
```

3. LIKE Operator with Pattern Matching

Definition:
The LIKE operator is used to search values that match a specific pattern.

Real-time example:
If you want to find all customers who live in cities starting with letter M (like Mumbai, Madurai).

Example Query:

```
SELECT *
FROM Client_Master
WHERE City LIKE 'M%';
```

4. DROP TABLE Command

Definition:
DROP TABLE is used to permanently delete a table and all its data from the database.

Real-time example:
If a table is created by mistake or no longer required, it can be removed completely.

Example Query:

```
DROP TABLE Client_Master;
```

5. Full Outer Join

Definition:
A FULL OUTER JOIN returns all records from both tables, including matched and unmatched rows.

Real-time example:
Consider two tables: one for customers and one for orders. Some customers may not have orders, and some orders may not be linked to customers.

Example Query:

```
SELECT *
FROM Customers C
FULL OUTER JOIN Orders O
ON C.customer_id = O.customer_id;
```

Write queries for following descriptions: (Joins)

1. Find out the products, which have been sold to 'Ivan Bayross'.

```
SELECT p.Description
FROM Client_Master c
JOIN Sales_Order s ON c.ClientNo = s.ClientNo
JOIN Sales_Order_Details d ON s.OrderNo = d.OrderNo
JOIN Product_Master p ON d.ProductNo = p.ProductNo
WHERE c.Name = 'Ivan Bayross';
```

Results | Messages

	Description
1	Trousers
2	Pull Overs
3	Pull Overs
4	Trousers
5	Pull Overs
6	Pull Overs

2. Finding out the products and their quantities that will have to be delivered in the current month.

```
SELECT p.Description, d.QtyOrdered
FROM Sales_Order s
JOIN Sales_Order_Details d ON s.OrderNo = d.OrderNo
JOIN Product_Master p ON d.ProductNo = p.ProductNo
WHERE MONTH(s.DelyDate) = MONTH(GETDATE())
  AND YEAR(s.DelyDate) = YEAR(GETDATE());
```

Results | Messages

	Description	QtyOrdered
1	Trousers	3
2	Pull Overs	2
3	Trousers	6
4	Shirts	1
5	Pull Overs	3
6	Trousers	3
7	Pull Overs	2
8	Trousers	6
9	Shirts	1
10	Pull Overs	3

3. Listing the ProductNo and description of constantly sold (i.e. rapidly moving) products.

SELECT p.ProductNo, p.Description
FROM Product_Master p
JOIN Sales_Order_Details d ON p.ProductNo = d.ProductNo
GROUP BY p.ProductNo, p.Description
HAVING COUNT(d.ProductNo) > 1;

Results | Messages

	ProductNo	Description
1	P00001	Trousers
2	P00002	Pull Overs
3	P00003	Shirts
4	P00004	Jeans
5	P00005	Jackets

4. Finding the names of clients who have purchased 'Trousers'.

SELECT DISTINCT c.Name
FROM Client_Master c
JOIN Sales_Order s ON c.ClientNo = s.ClientNo
JOIN Sales_Order_Details d ON s.OrderNo = d.OrderNo
JOIN Product_Master p ON d.ProductNo = p.ProductNo
WHERE p.Description = 'Trousers';

Results | Messages

	Name
1	Ivan Bayross
2	Sanjay

5. Listing the products and orders from customers who have ordered less than 5 units of 'Pull Overs'.

SELECT s.OrderNo, p.Description, d.QtyOrdered
FROM Sales_Order s
JOIN Sales_Order_Details d ON s.OrderNo = d.OrderNo
JOIN Product_Master p ON d.ProductNo = p.ProductNo
WHERE p.Description = 'Pull Overs'
  AND d.QtyOrdered < 5;

Results | Messages

	OrderNo	Description	QtyOrdered
1	O19001	Pull Overs	2
2	O19004	Pull Overs	3
3	O19001	Pull Overs	2
4	O19004	Pull Overs	3

Write queries for following descriptions: (Subqueries)

1. Finding the non-moving products i.e. products not being sold.

```
SELECT *
FROM Product_Master
WHERE ProductNo NOT IN (
    SELECT ProductNo FROM Sales_Order_Details
);
```

Results | Messages

	ProductNo	Description	ProfitPercent	UnitMeasure	QtyOnHand	ReorderLevel	SellPrice	CostPrice
1	P00006	Caps	5.00	Piece	200	50	800.00	700.00

2. Finding the name and complete address for the customer who has placed Order number 'O19001'.

```
SELECT Name, Address1, Address2, City, State, Pincode
FROM Client_Master
WHERE ClientNo = (
    SELECT ClientNo
    FROM Sales_Order
    WHERE OrderNo = 'O19001'
);
```

Results | Messages

	Name	Address1	Address2	City	State	Pincode
1	Ivan Bayross	Andheri	NULL	Mumbai	Maharashtra	400053

3. Finding the clients who have placed orders before the month of May'02.

```
SELECT *
FROM Client_Master
WHERE ClientNo IN (
    SELECT ClientNo
    FROM Sales_Order
    WHERE OrderDate < '2002-05-02'
);
```

Results | Messages

	ClientNo	Name	Address1	Address2	City	Pincode	State	BalDue
1	C00007	Ramesh	Ashok Nagar	NULL	Delhi	110001	Delhi	4000.00
2	C00008	Suresh	Alkapuri	NULL	Vadodara	390007	Gujarat	6000.00
3	C00009	Mahesh	Navrangpura	NULL	Ahmedabad	380009	Gujarat	3500.00
4	C00010	Kiran	Banashankari	NULL	Bangalore	560070	Karnataka	7200.00
5	C00011	Prakash	Karve Nagar	NULL	Pune	411052	Maharashtra	9100.00

Write Commands to do following

1. Display system date is Saturday, February 11, 2012.

SELECT FORMAT(CAST('2012-02-11' AS DATE),'dddd, MMMM dd, yyyy') AS SystemDate;

Results | Messages

	SystemDate
1	Saturday, February 11, 2012

2. Display Balance Due from Client master as $99,999.99.

SELECT FORMAT(BalDue,'$99,999.99') AS BalanceDue
FROM Client_Master;

Results | Messages

	BalanceDue
1	$999995000099
2	$999998000099
3	$999993000099
4	$9999912000099
5	$999992000099
6	$999999000099

3. Display message as 'Salesman Aman sold 50 goods of 50 while the given target was 100.'

SELECT 'Salesman Aman sold goods of 50 while given target was 100.' AS Message;

Results | Messages

	Message
1	Salesman Aman sold goods of 50 while given targ...

4. Display your Age in Years.

SELECT DATEDIFF(YEAR,'2002-06-15',GETDATE()) AS Age;

Results | Messages

	Age
1	21